Lesson 5.8 Guided Notes

Session 1 · Surface Area of Pyramids

I can find the surface area of a pyramid by adding the base area and the areas of its triangular faces. · 6.G.A.4

Word / Spanish	Meaning — complete the blank	Example
slant height / altura inclinada	The height of a side triangle, measured _____ its face.	½ × base × slant height
lateral face / cara lateral	A side _____ of a pyramid.	4 on a square pyramid
apex / ápice	The _____ point where the side faces meet.	tip of the pyramid

Key idea

SA of a pyramid = base area + area of all the _____ faces.

A square pyramid has _____ matching triangles.

Use the _____ height, not the straight-up height, for the triangles.

Worked example

A square pyramid has a base 10 cm on each side and a slant height of 7 cm. Find its surface area.

Base: 10 × 10 = _____ cm².

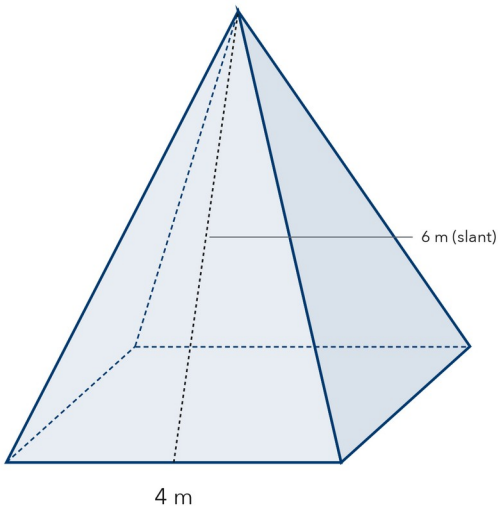
One triangle: ½ × 10 × 7 = _____ cm².

Four triangles: 4 × 35 = _____ cm².

Answer: SA = 100 + 140 = _____ cm²

You try

1. Find the surface area of the square pyramid.



_____

2. Find each area.
a. one triangle, base 6 in, slant 5 in _____ in²
b. square base, side 6 in _____ in²

_____

Sum it up: To find the surface area of a pyramid, I add ___ and ___.

_____

Session 2 · Lateral Area of Pyramids

I can find the lateral area of a pyramid, which covers only its slanted faces. · 6.G.A.4

Word / Spanish	Meaning — complete the blank	Example
lateral area / área lateral	The area of the slanted faces _____.	the base is left out
surface area / área superficial	The area of _____ face, base included.	lateral area + base
lateral face / cara lateral	A _____ side triangle of a pyramid.	3 on a triangle-base pyramid

Key idea

Lateral area = the area of all the _____ faces.

Surface area = lateral area + _____ area.

Read the problem to decide whether the base is _____.

Worked example

A square pyramid roof has a base 4 m on each side and a slant height of 6 m. Shingles cover only the slanted sides.

Cover the 4 slanted triangles. Leave out the _____.

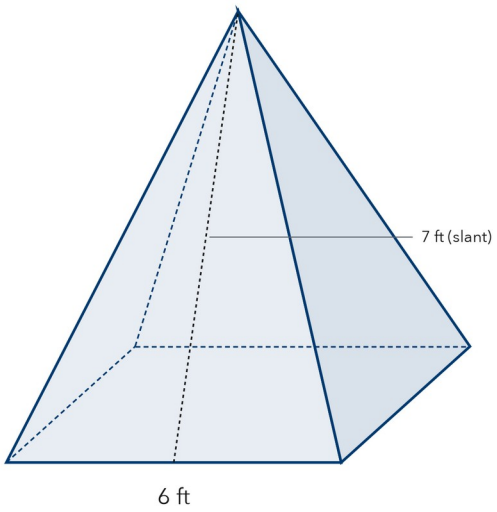
One triangle: ½ × 4 × 6 = _____ m².

Four triangles: 4 × 12 = _____ m².

Answer: lateral area = _____ m²

You try

1. Find the lateral area of the square pyramid.



_____

2. Find each lateral area.

a. SA = 90 m², base = 25 m² _____ m²

b. 3 faces, each 12 in² _____ in²

_____

Sum it up: I leave the base out when ___ because ___.

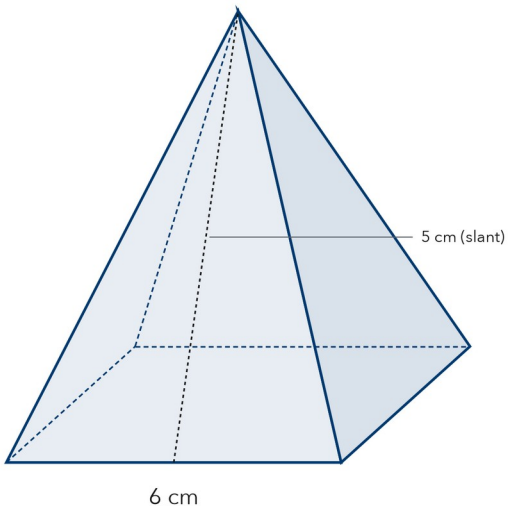
_____

Session 1 · Model, reason, and explain

Surface Area of Pyramids · 6.G.A.4

Success: represent the quantities, justify the relationship, and check the result with the correct units.

A square pyramid has base edge 6 cm and slant height 5 cm. Draw its net and distinguish slant height from vertical height. Find surface area.



_____

_____

_____

Extend your reasoning

Create a contrasting example that exposes the misconception: uses the vertical height for the triangular face area.. Explain what changes and what stays true.

_____

_____

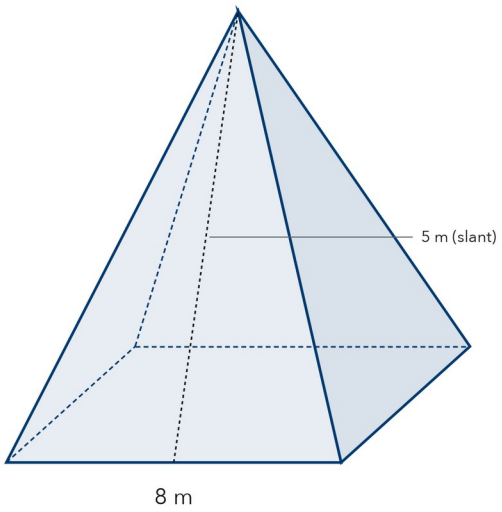
Self-check: Did my model, calculation, explanation, and units agree?

Session 2 · Model, reason, and explain

Lateral Area of Pyramids · 6.G.A.4

Success: represent the quantities, justify the relationship, and check the result with the correct units.

A square pyramid roof has base edge 8 m and slant height 5 m. Only its slanted faces need covering. Find the covered area and compare with the complete surface area.



_____

_____

_____

Extend your reasoning

Create a contrasting example that exposes the misconception: includes the base even when the context excludes it.. Explain what changes and what stays true.

_____

_____

Self-check: Did my model, calculation, explanation, and units agree?